RAY – AI Automation Engineer Case Study

Benjamin Kariuki · November 2026

Code repository	github.com/benjaminkariuki/ray-ai-automation-case-study
Task 1 – Google Sheet	docs.google.com/spreadsheets/d/1GwAt8R3_O-h6ivZ_CT_8yD6GMIdybyf2-YgscTQ9hRI
Task 1 – n8n workflow export	task1-job-outreach/workflow/job-outreach-workflow.json
Task 1 – README / user guide	task1-job-outreach/README.md
Task 1 – Sample output	task1-job-outreach/sample-output (CSV + JSON)
Task 2 – Notion	Added in the next section

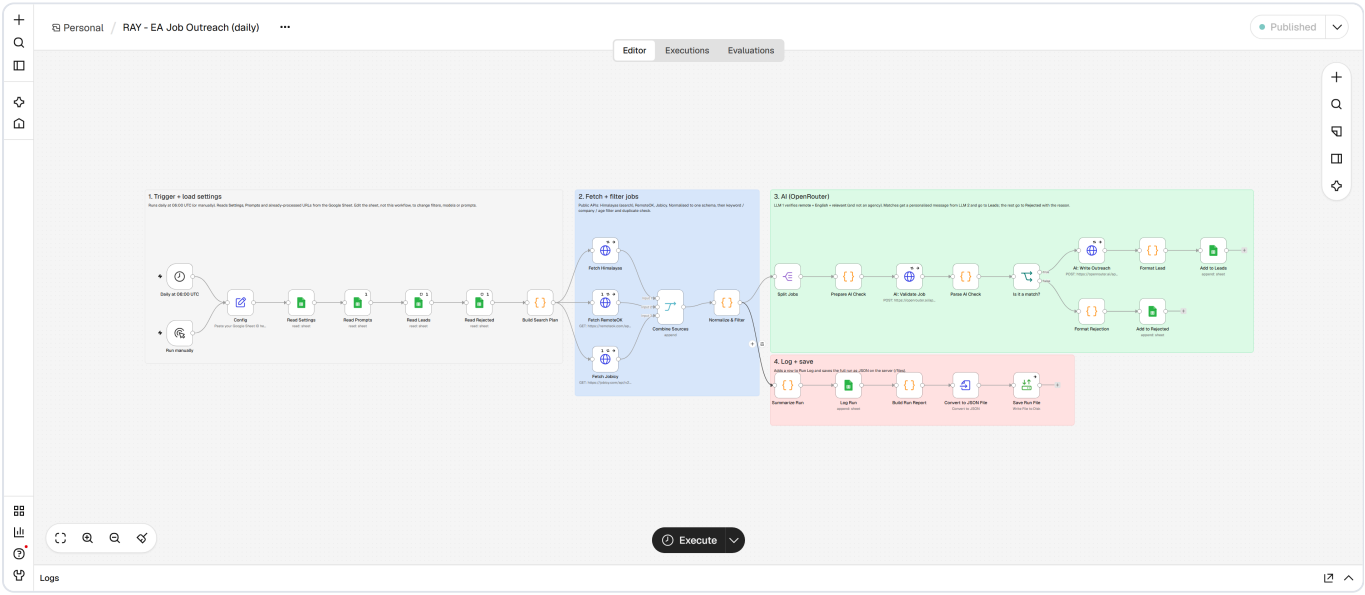
Task 1 – Executive Assistant Job Outreach Automation

What I built

I built an **n8n workflow**, self-hosted on Azure with HTTPS. It runs **every day at 06:00 UTC**:

- It pulls Executive Assistant and Admin Assistant postings from the public APIs of three job boards: **Himalayas, RemoteOK and Jobicy**. LinkedIn is not used.
- An LLM (via **OpenRouter**, free models) checks that each posting is **remote, English-speaking and relevant**, and that it isn't a staffing agency.
- A second LLM call writes a **personalised outreach email** for each match.
- Everything goes into a **Google Sheet that sales reps use as their dashboard**.

Filters, AI models and prompts are all **edited in the Sheet**, so tone can be tested without touching the workflow.



The n8n workflow (33 nodes, 4 colour-coded stages): trigger and settings, fetch and filter, AI check and outreach, log and save.

How each requirement is met

Requirement	How it's met
✓ Compliant, non-LinkedIn source	Three official public APIs (Himalayas, RemoteOK, Jobicy), with no HTML scraping. Each lead links to the original posting and names its source, as the boards' terms ask.
✓ Fields collected and stored	Job Title, Company, Location, Description, URL and Language Requirements, plus Source and Date Posted. Stored in the Sheet and as a **JSON file per run**; CSV exports are in the repo.
✓ Filter and validate (all 3)	A keyword, company and age pre-filter runs first. Then the LLM returns strict JSON: `is_remote`, `is_english`, `is_relevant`, `is_agency` and a reason. Only jobs passing all three checks are kept. Rejections are logged with the reason so they can be audited.
✓ Personalised LLM message	Each email cites specific duties or tools from that posting. The prompt allows facts about the company only from the posting, and facts about us only from the "About us" setting.
✓ Interchangeable prompt	Prompts live in the Sheet's **Prompts** tab. Changing `Settings → outreach_prompt` (for example `v1_warm → v2_direct`) switches the tone on the next run, and each lead records the **Prompt Used**.
✓ Google Sheets columns	The Leads tab has the 6 required columns in order, then Language, Source, Dates, Prompt Used, **Status** (a dropdown) and Notes.
✓ Scheduling	An n8n Schedule Trigger runs daily at 06:00 UTC on an always-on Azure VM. There's also a manual Execute button.
✓ README / user guide	Covers install (existing n8n, or one-command Azure deploy), running, changing filters and prompts, troubleshooting, and a guide for sales reps. A short guide is also built into the Sheet.

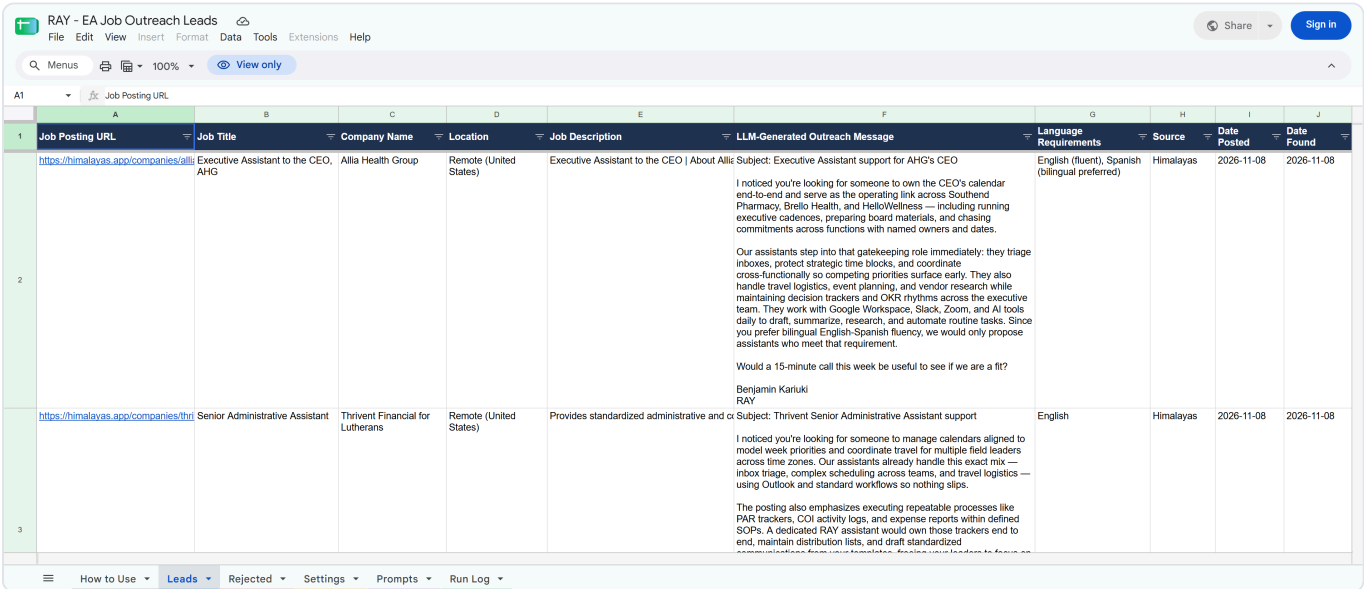
Results from live test runs

264–359
postings fetched per run

~55
pass the keyword filter

15
new jobs checked by AI per run (configurable)

10 leads
written in one run (GitLab, Pearson, Thrivent …)



The Leads tab in the Google Sheet: required columns, the AI-written message, and a Status dropdown for the sales rep.

Example of a generated message

> Subject: GitLab Office of the CFO support
>
> I noticed you're looking for an Executive Assistant to support multiple Finance leaders across time zones, manage travel and expenses in Navan, and maintain shared Office of the CFO resources in Slack and Google Workspace. That mix of calendar complexity, policy-driven expense reconciliation, and cross-functional coordination is exactly what our assistants handle daily. …
>
> Would a 15-minute call this week be useful to see if we are a fit?

Key design decisions

- **Cheap filter first, AI second.** Keyword rules remove about 85% of postings before any LLM call, so the workflow runs inside OpenRouter's free tier. A per-run cap protects the quota.
- **Agency detection.** Many remote EA postings come from VA and staffing agencies, which are competitors, not leads. They're excluded both by name and by the AI.
- **Reliability.** Jobs already in the Sheet are skipped, so there are no duplicates and no repeat AI cost. If an AI call fails (rate limit, outage), the job isn't saved and is retried automatically on the next run. A fallback model takes over when the main model is busy, and each job board fails independently of the others.
- **Transparency.** The **Rejected** tab shows Yes/No for each check with the AI's reason. The **Run Log** tab shows how many jobs were fetched, filtered, checked, added and errored on each run.
- **Editable by non-technical users.** Search terms, filters, models, sender details and prompts are all Sheet cells.

Process notes

Code-node JavaScript is kept as readable files in `workflow/code/` and assembled into the importable workflow by `build_workflow.py`. `setup_sheet.py` builds the formatted Sheet in one command. While testing, I tightened the prompts after the model invented details: once a language skill, once that our assistants already knew a client's tools. I also clarified that a time-zone requirement still counts as remote. Free-tier models are occasionally rate-limited. The Run Log records this, and affected jobs are retried automatically.

Task 2 – Executive Travel Planner (Notion)

To be completed.